

The Effect of the Virtual Object Size on Weight Perception Augmented with Pseudo-Haptic Feedback

Jinwook Kim*

Jeongmi Lee†

Graduate School of Culture Technology
Korea Advanced Institute of Science and Technology

ABSTRACT

Providing realistic haptic feedback of virtual objects is critical for immersive VR experience, and there have been many approaches using proprioception mismatch to provide pseudo-haptic feedback. Most of them, however, are limited in that the differences in visual features of the objects are not considered in adjusting the level of pseudo-haptic feedback. Therefore, we conducted an experiment to examine how the threshold of simulated weight perception changes according to the size of a virtual object. The results showed that a stronger level of visual pseudo-haptic feedback is required for bigger objects for simulated weight perception.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

The development of head-mounted display (HMD) devices for virtual reality (VR) has increased accessibility and usability of VR for the general public. However, there are still many issues left to be solved for VR to provide a more immersive and realistic experience to users. Companies and researchers are taking various approaches to adopt real-world experience in the virtual environment. The most common method used to augment the sense of real-world is pseudo-haptic feedback [1].

Pseudo-haptic feedback with visual stimuli has been studied and adopted in various devices and applications (e.g., gamepad, joystick, and mobile phone) for a long time [1–3]. For instance, visual manipulation for simulating the sense of friction and stiffness of virtual objects was widely used in games and 2D user interfaces. Since VR has become the emerging technology nowadays, methods of assigning the sense of real-world objects (e.g., weight) on virtual objects have been actively studied. Some of these studies used the height difference between the real hand and the virtual hand (Control/Display ratio) to induce a weight perceptual illusion [4, 5]. In these studies, the acceptance range of the C/D ratio that successfully induced the weight perception was measured to provide design guidelines for simulating weight perception in VR. In addition, Rietzler et al. adopted their pseudo-haptic feedback model on the bowling game and found that participants experienced a higher level of presence and immersion with the pseudo-haptic feedback [4]. However, the pre-existing pseudo-haptic feedback approaches that manipulate the C/D ratio have limitations in that the visual differences between virtual objects are not considered in determining the level of pseudo-haptic feedback. Since the visual as well as somatosensory information is used to perceive the weight of real-world objects [6], it is critical for the high fidelity of simulated weight perception to understand how visual features (e.g., size,

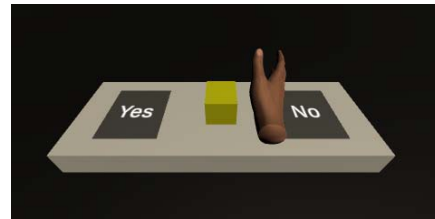


Figure 1: The experiment environment in virtual reality. Participants were asked to lift the yellow cube and respond by placing the cube on the Yes or No area based on weight perception.

shape, and color) of a virtual object influence weight perception and to adjust the level of pseudo-haptic feedback accordingly.

Therefore, in this study, we examined the relationship between the sense of weight simulated with pseudo-haptic feedback and the size of a virtual object. We independently manipulated the C/D ratio and the size of a virtual object, and measured the absolute threshold of weight perception in each condition to probe how users' simulated weight perception is influenced by the object size and the C/D ratio.

2 IMPLEMENTATION

We used the Oculus Rift S (2560x1440 resolution, 115-degree FoV, 80Hz) and Unity to implement the pseudo-haptic feedback and virtual environment. As in Figure 1, a yellow cube was placed on the center of a table in virtual reality, with the size of the cube selected among three different values (10, 15, and 20 cm; see Figure 2). On each trial, participants were asked to manipulate the right virtual hand with a controller to grab and lift the cube, and then place the cube on one of the response (Yes or No) area based on whether they felt the weight of the cube or not. Only while participants were grabbing the virtual cube, a temporal offset was applied for visual pseudo-haptic feedback, by using the linear interpolation on the target coordination by each frame. A C/D ratio between 0 to 1, randomly selected from six different levels (i.e., 0.05, 0.15, 0.25, 0.35, 0.5 and 1), was applied to the virtual hand and the cube to make the visual display of the movement *slower* than reality; the C/D ratio level of 1 indicates no pseudo-haptic feedback, and as the value gets closer to 0, the movement of the hand and the object becomes *slower* than the real motion. We did not include the left hand in the VR environment and did not apply the temporal offset effect on the cube falling event to minimize the effects of other confounding variables and to focus on the effect of pseudo-haptic feedback.

3 EXPERIMENT

We hypothesized that the effect of visual pseudo-haptic feedback on simulated weight perception would occur differently based on the size of the virtual cube. Thus, using pseudo-haptic feedback, we conducted an experiment to measure and compare the absolute thresholds of weight perception for three differently-sized cubes. The details of the experiment are described as follows.

*e-mail: jinwook.kim31@kaist.ac.kr

†Corresponding author. e-mail: jeongmi@kaist.ac.kr

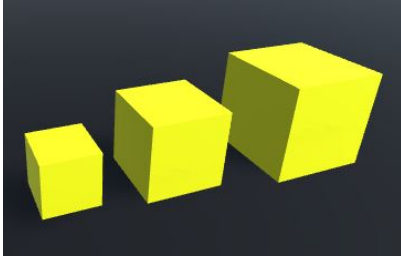


Figure 2: Illustration of virtual cubes in three different sizes used in the experiment. The length of the side of the smallest cube on the left was 10 cm, followed by 15 and 20 cm.

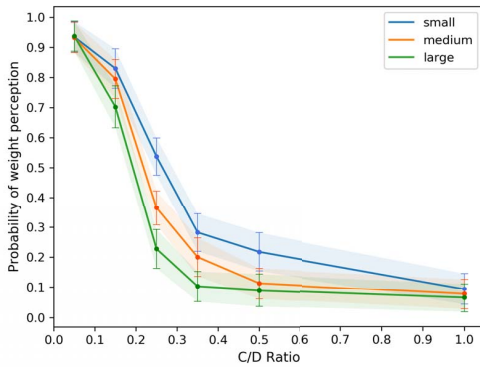


Figure 3: Participants' responses of weight perception of the virtual object. The average probability of responding "Yes" to the weight perception question is plotted for each C/D ratio and size condition. Error bars represent standard error of the mean.

3.1 Participants

A total of 20 students (8 male; mean age 25.25 (SD=2.77); 3 left-handed; all had experience in VR) from a university participated in the experiment and received a \$10 gift card. All of the study protocols and methods were approved by the KAIST Institutional Review Board (IRB), and all participants provided written informed consent.

3.2 Procedure

The experimenter first explained about the task and the procedures. On each trial, participants were instructed to lift the cube up and place it on the Yes or No area (see Figure 1), depending on their perception of weight of the virtual cube. The experiment contained a total of 360 trials, consisting of six levels of C/D ratio and three different cube sizes, repeated 20 times each. The order of trials was randomized for each participant.

3.3 Results

We used two-way repeated measures ANOVA to analyze the effects of C/D ratio and cube size on weight perception responses (see Figure 3). The results showed that there was a significant main effect of C/D ratio on weight perception ($F(1.56, 29.72)=47.39, p<.001$, Greenhouse-Geisser corrected). The post-hoc comparisons indicated that the probability of perceiving weight was higher for C/D ratios under 0.15 ($M=85.5\%$, $SD=0.27$) as compared to those over 0.35 ($M=13.8\%$, $SD=0.25$). Importantly, the main effect of cube

size was significant ($F(1.11, 23.22)=22.68, p<.001$, Greenhouse-Geisser corrected), with the higher mean probability of simulated weight perception as the size of the cube got smaller (Small: 48.3% ($SD=0.4$); Medium: 41.4% ($SD=0.41$); Large: 35.4% ($SD=0.42$)). This indicates that participants were overall more likely to perceive weight induced by pseudo-haptic feedback when the virtual cube was smaller. Consistent with this result, the C/D ratio values corresponding to the absolute threshold (50% weight perception probability) were significantly different across cube sizes, namely 0.18, 0.21, and 0.26 for the large, medium, and small cube, respectively ($ps<.05$). Lastly, there was also an interaction between C/D ratio and cube size ($F(3.25, 61.84)=9.03, p<.001$, Greenhouse-Geisser corrected). The paired t-tests showed that the effect of cube size was significant within the intermediate C/D ratio range between 0.15 and 0.5, but not at the two extreme levels (0.05 and 1). This could be explained by a ceiling effect and a floor effect.

4 DISCUSSION & CONCLUSION

In this study, we conducted an experiment measuring and comparing the absolute thresholds of simulated weight perception for differently sized virtual objects. We utilized different levels of C/D ratio to provide visual pseudo-haptic feedback. The results of the experiment showed that the absolute thresholds of simulated weight perception differed according to the cube size, which indicates that the effect of visual pseudo-haptic feedback occurs differently according to the size of the virtual object. As the cube size gets bigger, a lower C/D ratio (slower motion) is required for participants to barely perceive weight (50%). These results suggest that when designing weight on virtual objects with pseudo-haptic feedback, the level of C/D ratio needs to be adjusted depending on the size of the targeted object.

There are several limitations to the current study. We only investigated the effect of size on simulated weight perception, but there are other visual features that influence weight perception, such as shape and color. In future studies, the effects of various other visual features of objects on simulated weight perception should be studied. This line of research will contribute to constructing a more precise and effective model for designing weight on virtual objects in VR contents.

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